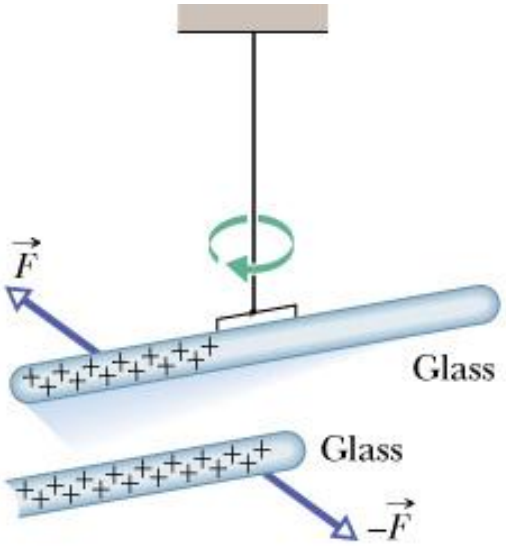
Chapter 21

Electric Charge

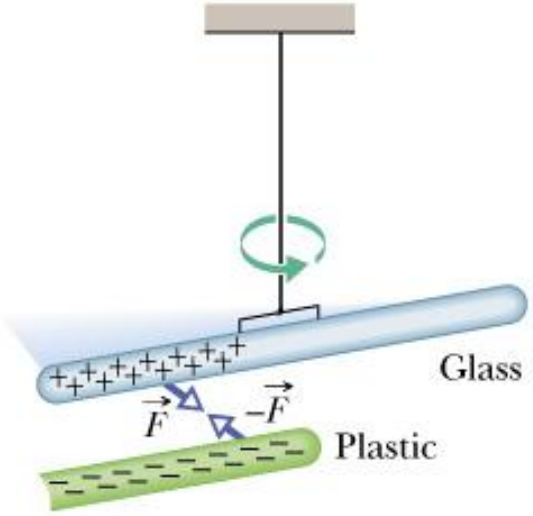
In this chapter we will introduce a new property of matter known as "electric charge" (symbol *q*).
We will explore the charge of atomic constituents.

Moreover, we will describe the following properties of charge:

- Types of electric charge
- Forces among two charges (Coulomb's law)
- Charge quantization
- Charge conservation



(a)

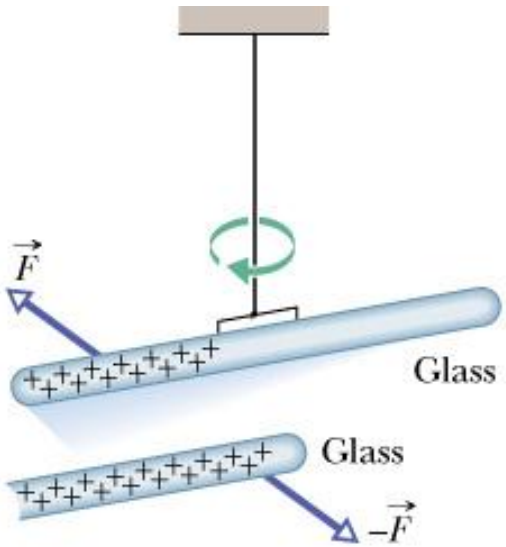


(b)

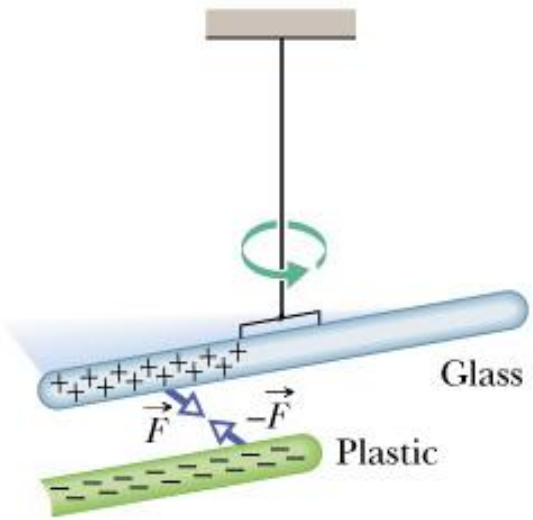
Empirically it was known since ancient times that if amber is rubbed on cloth, it acquires the property of attracting light objects such as feathers. This phenomenon was attributed to a new property of matter called "electric charge." (Electron is the Greek name for amber.) More experiments show that there are two distinct types of electric charge: positive (color code: red) and negative (color code: black). The names "positive" and "negative" were given by Benjamin Franklin.

When we rub a glass rod with silk cloth, both objects acquire electric charge. The sign on the charge on the glass rod is defined as **positive.**

In a similar fashion, when we rub a plastic rod with fur both objects acquire electric charge. The sign on the charge on the plastic rod is defined as **negative.**



(*a*)



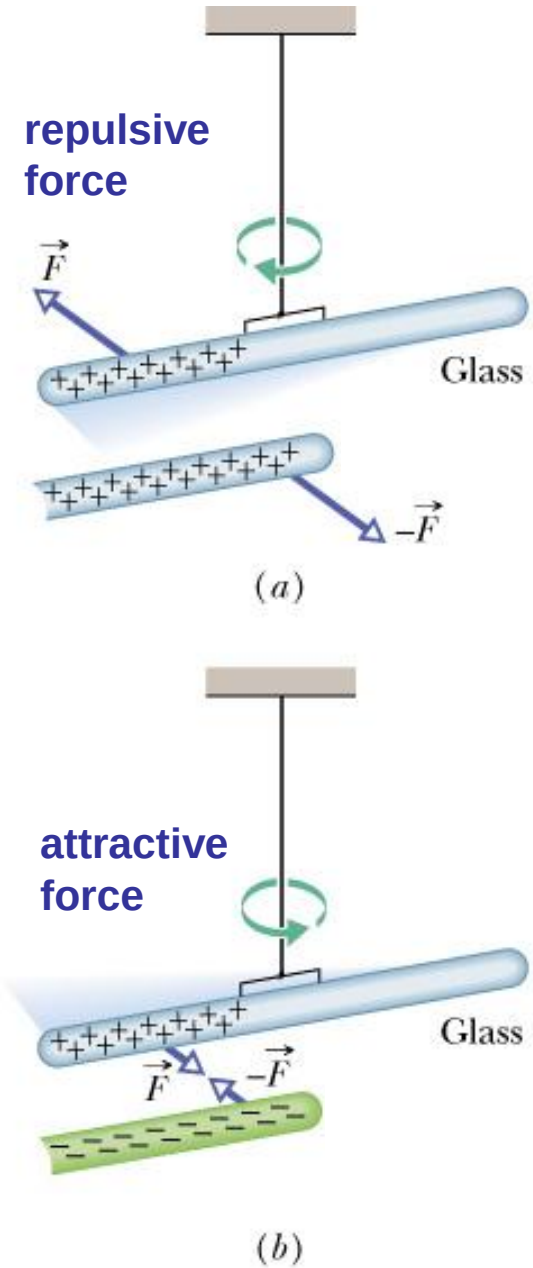
(*b*)

Q: Do we have enough information so as to be able to determine the sign of all other charges in nature? To answer this question we need one more piece of information.

Further experiments on charged objects showed that:

1. Charges of the same type (either both positive or both negative) repel each other (fig. *a*).
2. Charges of opposite type on the other hand attract each other (fig. *b*).
   The force direction allows us to determine the sign of an unknown electric charge.

Charges of the same sign repel each other. Charges of opposite sign attract each other.

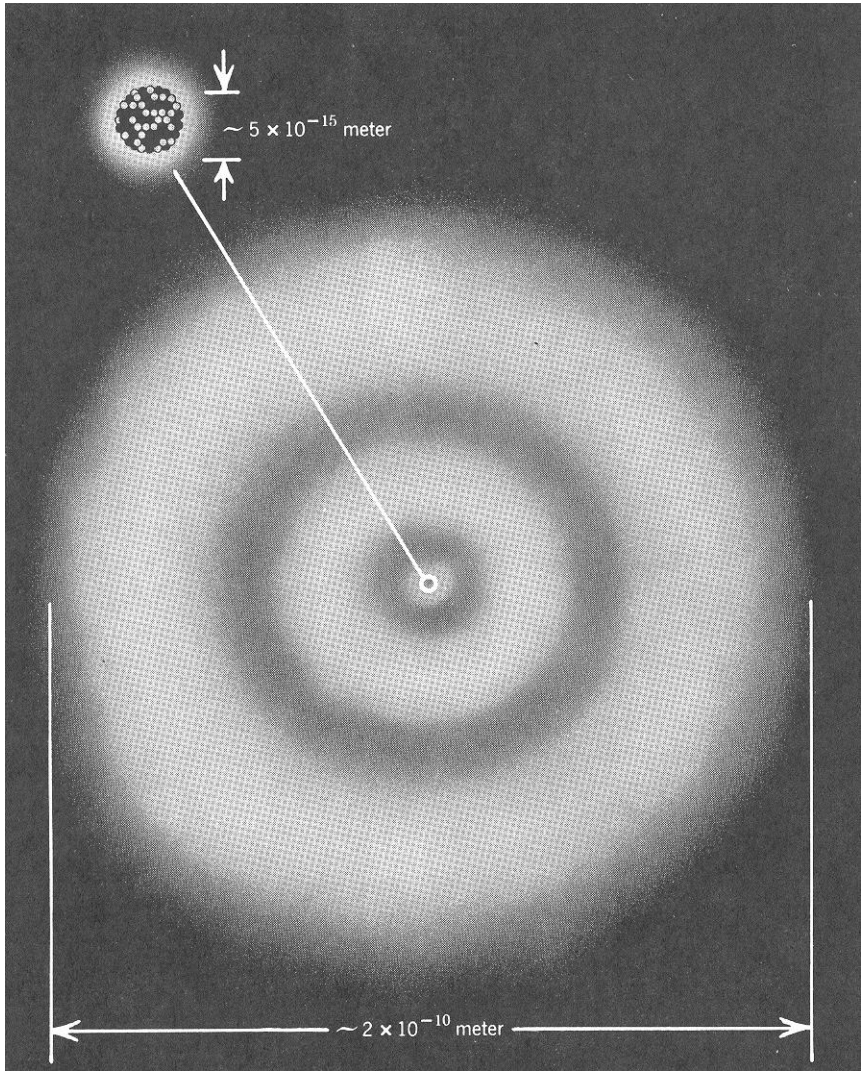


The recipe is as follows:

We charge a glass rod by rubbing it with silk cloth. Thus we know that the charge on the glass rod is positive. The rod is suspended in such a way so that it can keep its charge and also rotate freely under the influence of a force applied by charge with the unknown sign. We approach the suspended class rod with the new charge whose sign we wish to determine.
Two outcomes are possible. These are shown in the figure to the left:

Fig. *a*: The two objects repel each other. We then conclude that the unknown charge has a positive sign.

Fig. *b*: The two objects attract each other. We then conclude that the unknown charge has a negative sign.



In Benjamin Franklin's day (18th century) it was assumed that electric charge is some type of weightless continuous fluid. Investigations of the structure of atoms by Ernest Rutherford at the beginning of the 20th century revealed how matter is organized and also identified that charge of its constituents.

Atoms consist of **electrons** and the **nucleus**.
Atoms have sizes ~ 5×10^{-10} m.
Nuclei have sizes ~ 5×10^{-15} m.

The nucleus itself consists of two types of particles: protons and neutrons.
The electrons are **negatively** charged.
The protons are **positively** charged.
The neutrons are **neutral** (zero charge).

Thus electric charge is a fundamental property of the elementary particles (electrons, protons, neutrons) out of which atoms are made.

Mass and Charge of Atomic Constituents

Neutron (n) : Mass $m = 1.675\times10^{-27}$ kg; Charge $q = 0$

Proton (p) : Mass $m = 1.673\times10^{-27}$ kg; Charge $q = +1.602\times10^{-19}$ C

Electron (e) : Mass $m = 9.11\times10^{-31}$ kg; Charge $q = -1.602\times10^{-19}$ C

Note 1: We use the symbols "-e" and "+e" for the electron and proton charge, respectively. This is known as the **elementary charge.**

Note 2: Atoms are electrically neutral. The number of electrons is equal to the number of protons. This number is known as the " **atomic number** " (symbol: Z). The chemical properties of atoms are determined **exclusively** by Z.

Note 3: The sum of the number of protons and the number of neutrons is known as the " **mass number** " (symbol: A).

Notation: $^{235}_{92}\mathrm{U}$

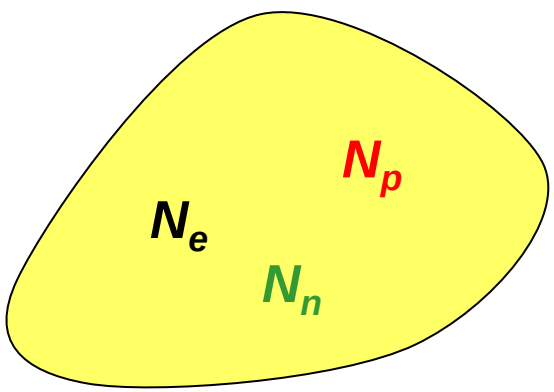
$Z = 92$ = number of protons/electrons
$A = 235$ = number of protons + neutrons

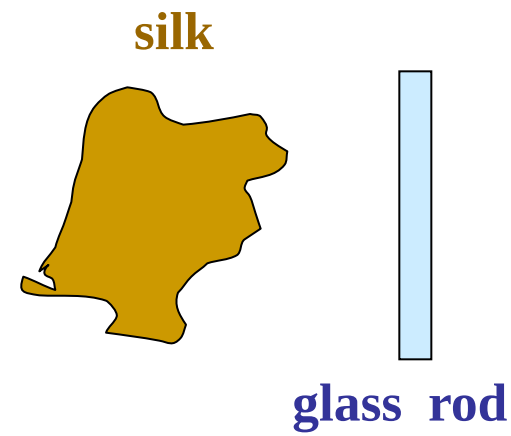
The atomic number Z = 92 defines the nucleus as that of a uranium atom.

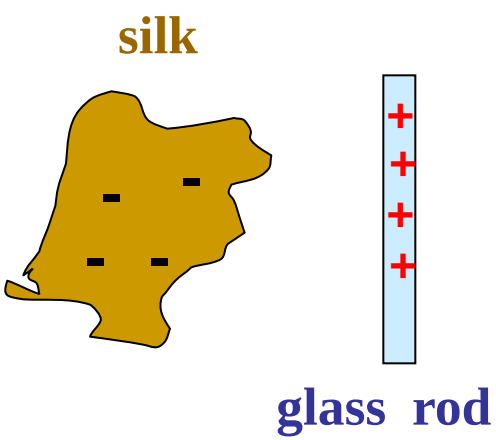
Charge Quantization

Now that we have identified the charge of the atomic constituents (electrons, protons, neutrons), it is clear that the net charge Q_{net} of an object that contains N_e electrons , N_p protons , and N_n neutrons is given by $Q_{\text{net}} = -eN_e + eN_p + 0N_n = e\left(N_p - N_e\right) = ne.$

Here $n = \left(N_p - N_e\right)$ and it is an integer. Thus the net charge is **quantized.** This means that it cannot take any arbitrary value but only values that are multiples of the elementary charge e. The value of e is small and thus in many large-scale phenomena the "graininess" of electric charge is not apparent.







Conservation of Charge

Consider a glass rod and a piece of silk cloth (both uncharged) shown in the upper figure. If we rub the glass rod with the silk cloth we know that positive charge appears on the rod (see lower figure). At the same time an equal amount of negative charge appears on the silk cloth, so that the net rod-cloth charge is actually zero. This suggests that rubbing does not create charge but only transfers it from one body to the other, thus upsetting the electrical neutrality of each body. Charge conservation can be summarized as follows: In any process the charge at the beginning equals the charge at the end of the process.

Net charge before = Net charge after

$$Q_i = Q_f$$

(21-8)

No exceptions of charge conservation have been found.

For example, charge is conserved in nuclear reactions.

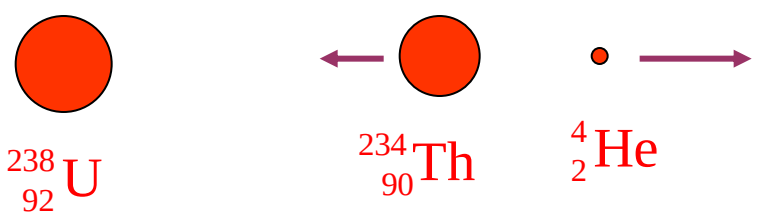
An example is given below:

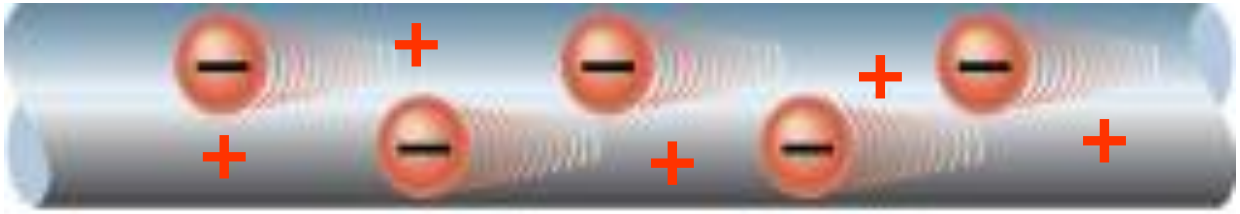
$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

In this example, a parent nucleus of Uranium-238, which has 92 protons and $(238\text{-}92) = 146$ neutrons, decays into two products:

i. A daughter Thorium-234 nucleus, which consists of 90 protons and $(234\text{-}90) = 144$ neutrons

ii. A Helium-4 nucleus, which has 2 protons and 2 neutrons. The net charge before and after the decay remains the same, equal to 92e.





Conductors and Insulators

Conductors are materials that allow charges to move freely through them.

Examples are: copper, aluminum, mercury.

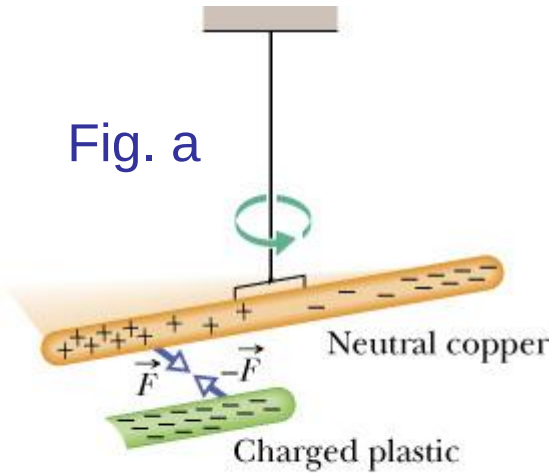
Insulators are materials through which charges cannot move freely.

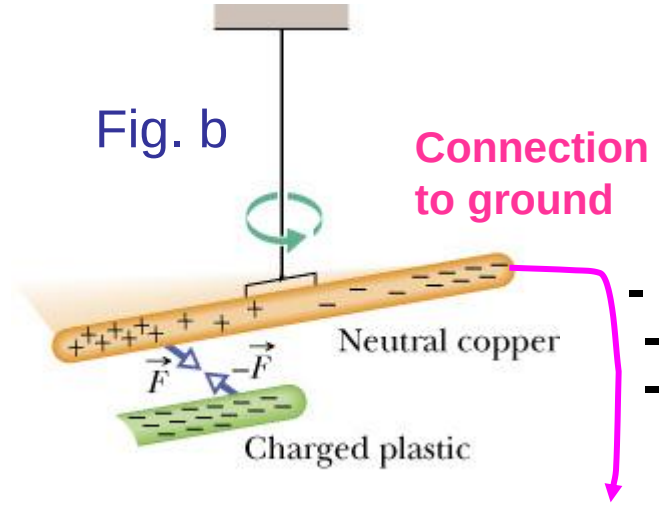
Examples are: plastic, rubber, glass, ceramics.

In conductors, one or more of the outermost electrons of the constituent atoms become free and move throughout the solid. These are known as conduction electrons. The conduction electrons leave behind positively charged atoms (known as ions). Only the negatively charged electrons are free to move inside a conductor. The positively charged ions are fixed in place.

Insulators do not have conduction electrons.

Charging a Conductor by Induction

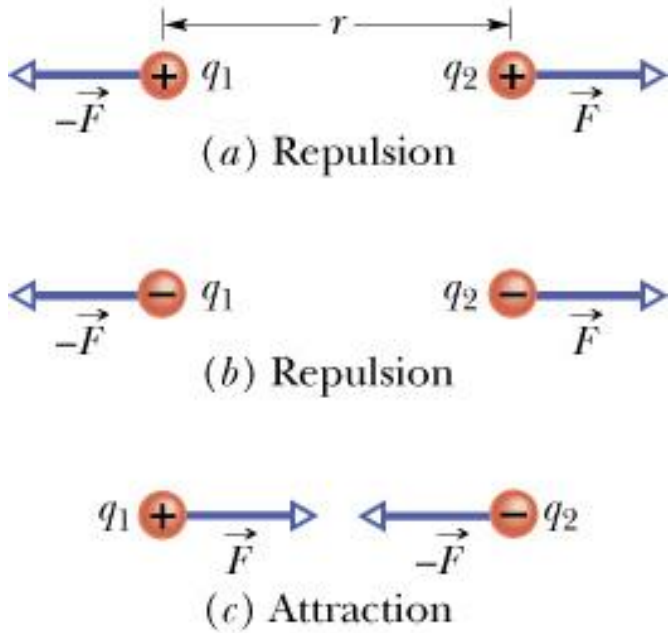




A conductor can be charged using the procedure shown in fig. *a* and fig. *b*. In fig. *a* a conductor is suspended using an insulating thread. The conductor is initially uncharged. We then approach the conductor with a negatively charged rod. The negative charges on the rod are fixed because plastic is an insulator. These repel the conduction electrons of the conductor, which end up at the right end of the rod. The right end of the rod has an electron deficiency and thus becomes positively charged. In fig. *b* we provide a conducting path to ground (e.g., we can touch the conductor). As a result, the electrons escape to the ground. If we remove the path to the ground and the plastic rod, the conductor remains positively charged.

Note 1 : The induced charge on the conductor has the opposite sign of the charge on the rod.

Note 2 : The plastic rod can be used repeatedly.



$$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2}$$

$$F = G\frac{m_1 m_2}{r^2}$$

(21-12)

Coulomb's Law

Consider two charges q_1 and q_2 placed at a distance r. The two charges exert a force on each other that has the following characteristics:

1. The force acts along the line connecting the two charges.

2. The force is attractive for charges of opposite sign. The force is repulsive for charges of the same sign.

3. The magnitude of the force, known as Coulomb force, is given by the equation $F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2}$.

The constant ε_0 is known as the **permittivity constant** $\varepsilon_0 = 8.85\times10^{-12}\ \mathrm{N\cdot m^2/C^2}$.

The Coulomb force has the same form as Newton's gravitational force. The two differ in one aspect: The gravitational force is always attractive. Coulomb's force on the other hand can be either attractive or repulsive depending on the sign of the charges involved.

Units of Charge

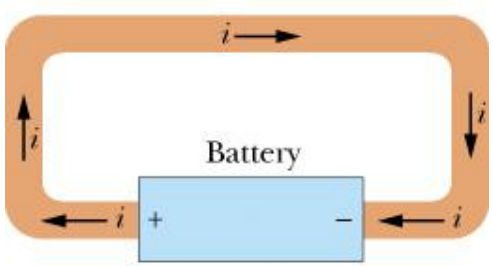
The unit of charge in the SI unit system is the "Coulomb" (symbol C).

In principle we could use Coulomb's law for two equal charges q as follows:

Place the two charges at a distance $r = 1m$. $q = 1$ C if $F = \dfrac{1}{4\pi\varepsilon_0} = 8.99\times10^9$ N:

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2} \rightarrow F = \frac{1}{4\times3.14\times8.85\times10^{-12}}\frac{1^2}{1^2} = 8.99\times10^9 \text{ N}$$

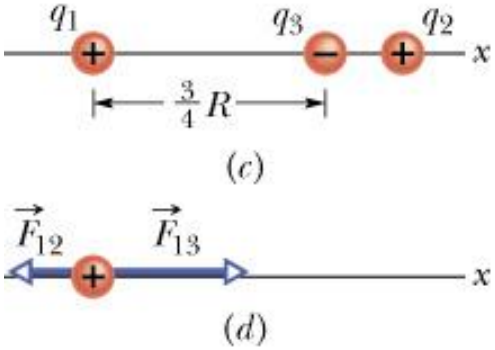
For practical reasons that have to do with the accuracy of the definition, the electric current is used instead. The electric current i in the circuit of the figure is defined by the equation $i = \dfrac{dq}{dt}$, i.e., the amount of charge that flows through any cross section of the wire per unit time. The unit of current in SI is the ampere (symbol A) and it can be defined very accurately.



(b)

If we solve the equation above for dq we get $dq = idt$. Thus if a current $i = 1$A flows through the circuit, a charge $q = 1$C passes through any cross section of the wire in one second.

(21-13)



Coulomb's Law and the Principle of Superposition

The net electric force exerted by a group of charges is equal to the vector sum of the contribution from each charge.

For example, the net force $\vec{F}_1$ exerted on q_1 by q_2 and q_3 is equal to $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$.

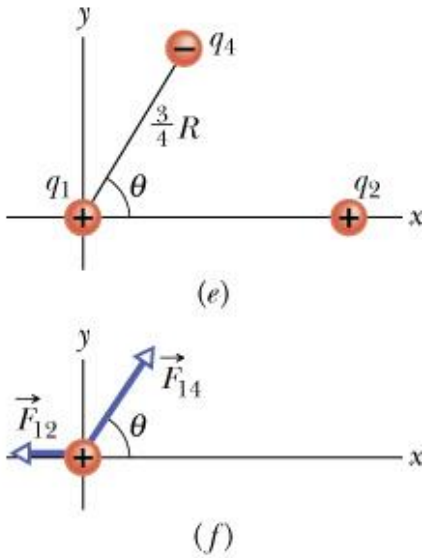
Here $\vec{F}_{12}$ and $\vec{F}_{13}$ are the forces exerted on q_1 by q_2 and q_3, respectively.

In general, the force exerted on q_1 by n charges is given by the equation

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \ldots + \vec{F}_{1n} = \sum_{i=2}^{n} \vec{F}_{1i}$$

One must remember that $\vec{F}_{12}$, $\vec{F}_{13}$, ... are vectors and thus we must use vector addition. In the example of fig. f we have:

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{14}$$



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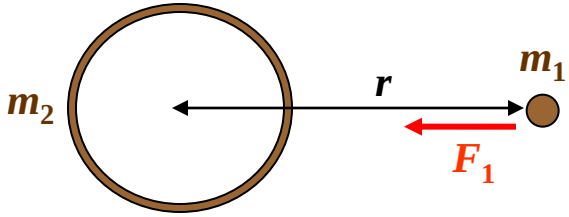
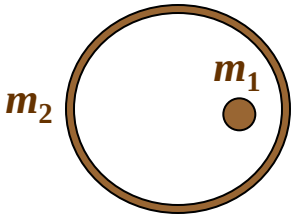
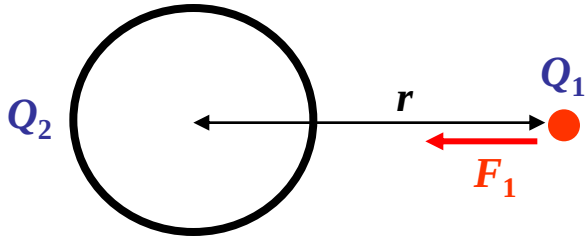


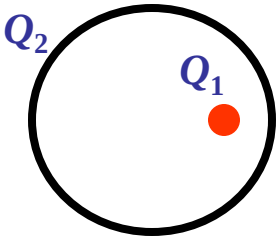
$$F_1 = G\frac{m_1 m_2}{r^2}$$



$$F = 0$$



$$F_1 = \frac{1}{4\pi\varepsilon_0}\frac{|Q_1||Q_2|}{r^2}$$



$$F_1 = 0$$

The gravitational force that a uniform shell of mass m_2 exerts on a particle of mass m_1 that is outside the shell is given by the the equation $F_1 = G\frac{m_1 m_2}{r^2}$.

It is as if the shell's mass m_2 were all concentrated at the shell center.

If m_1 is inside the shell, the net force exerted by m_2 is zero.

Because of the similarity between Newton's gravitational law and Coulomb's law, the same is true for the electric force exerted by a spherical shell of charge Q_2 on a point charge Q_1. If Q_1 is outside the shell, then the force F_1 exerted by Q_2 is

$F_1 = \frac{1}{4\pi\varepsilon_0}\frac{|Q_1||Q_2|}{r^2}$. If Q_1 is inside the shell, then the force $F_1 = 0$.

(21-15)